

AI-BASED REAL-TIME SIGN LANGUAGE TRANSLATION SYSTEM

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OBJECTIVES

This project studies sign language recognition from skeleton keypoints in two settings:

- Build a phase-aware **isolated** ASL recognizer and test whether phonological phase structure helps.
- Build a feature-space **continuous** (CSLR) system and test which modality wins, and whether fusion helps.
- Run every comparison under matched, frozen-feature, single-GPU conditions, reporting results honestly.

INTRODUCTION

Sign languages are complete natural languages, yet automatic recognition is hard in both canonical settings: **isolated** recognition (classifying one sign) and **continuous** recognition (transcribing an unsegmented stream into glosses).

Compact skeleton keypoints are an attractive input — signer-invariant and directly encoding the hand and body configurations that make up signing.

Our question: what *additional* structure improves recognition on top of keypoints, and do the gains survive controlled, low-compute evaluation rather than reflecting tuning?

DATA & TOOLS

- **ASL Citizen** (isolated, top-100 classes).
- **RWTH-PHOENIX-Weather 2014** (continuous; 5,672/540/629 splits).
- MediaPipe keypoints; frozen Kinetics-I3D video features; HRNet whole-body keypoints.
- PyTorch, OpenCV, NumPy; single free Google Colab GPU.

METHODS

Part I (isolated): MediaPipe keypoints → 444-d motion features → phase-detection TCN (prep/stroke/retraction) → recognition TCN with attention.

Part II (continuous): two frozen streams (I3D video + HRNet keypoints) → temporal encoders → concatenation fusion → BiLSTM → CTC training + greedy decoding.

CONCLUSIONS

Across both tasks, skeleton keypoints carry the dominant signal. Added structure helps only when it supplies genuinely complementary information at a compatible representation: phase features give a real but non-significant isolated gain, while a fused video stream wins in the continuous task once the temporal-resolution mismatch is resolved.

HIGHLIGHTS

- Two reproducible, compute-frugal pipelines (~12–15 GPU-hours total).
- A controlled modality study with a diagnosed-and-resolved negative result.
- A statistically rigorous null result and a real-time webcam demonstrator.

KEY RESULT

Skeleton keypoints beat generic video features by **8.66 WER points** (44.71% vs. 53.37%). Once a diagnosed temporal-resolution mismatch is fixed (stride 2), multimodal fusion reaches **40.61% test WER**, beating the best single stream by **4.10 points**.

WORKFLOW

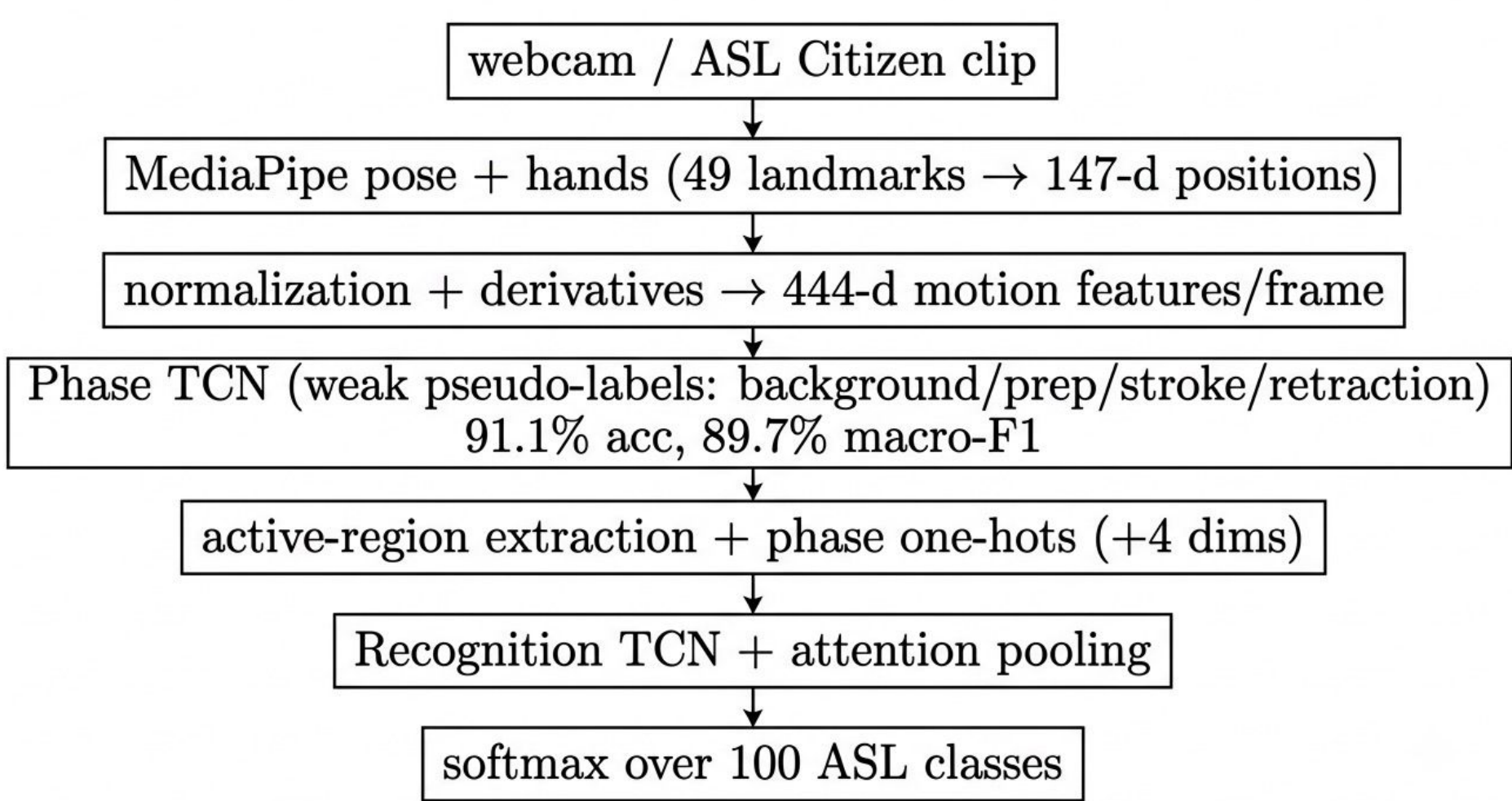


Figure 1: Part I as-built pipeline: phase-aware isolated ASL recognition.

RESULTS

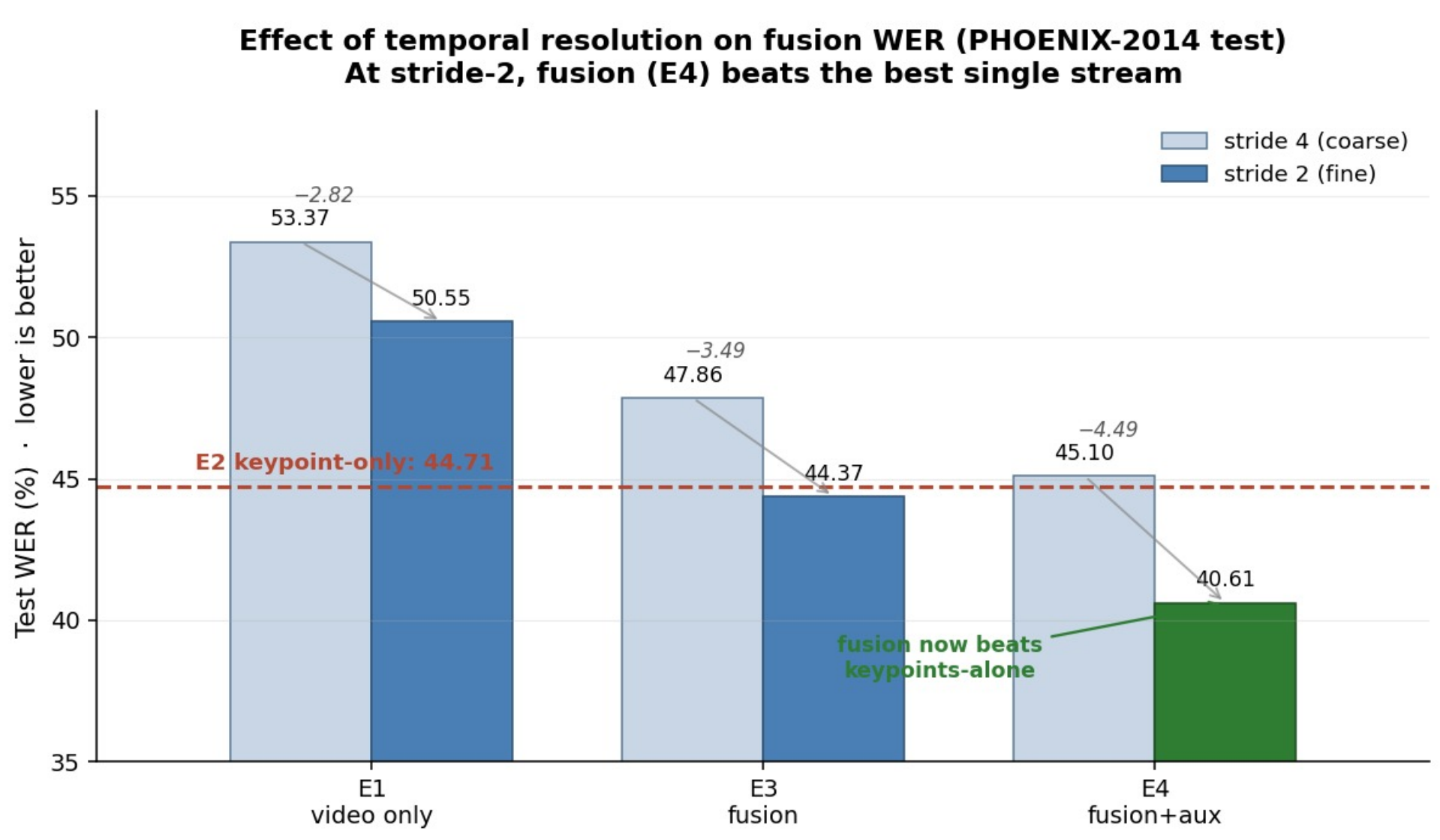


Figure 4:WER across configurations (stride 4 vs. stride 2).

Figure 2:Part I pipeline: phase-aware isolated ASL recognition.

Isolated phase features give a small, consistent but non-significant gain; continuous fusion wins decisively once stream resolutions match:

Model	Stride 4	Stride 2
Video-only	53.37	50.55
Keypoint-only	44.71	—
Fusion + aux. CTC	45.10	40.61

Table 1:Test WER (%) on PHOENIX-2014, lower is better

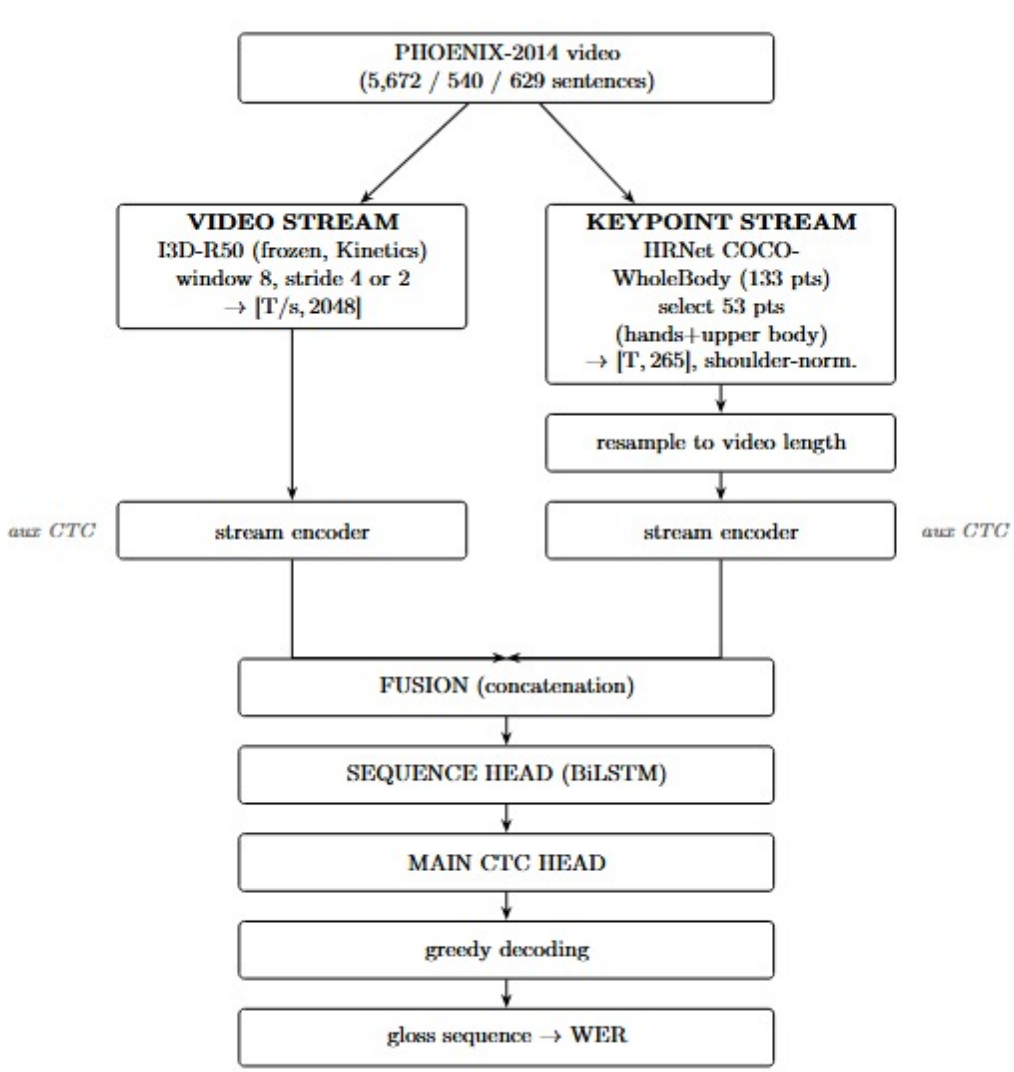


Figure 3:Part II architecture: multimodal CSLR system.

FUTURE WORK

- Deploy the recognizer as a real-time web service for live, in-browser sign recognition.
- Extend this study into a publishable research paper.
- Add beam-search decoding with a gloss language model to suppress CTC repeat-insertions.
- Replace the generic I3D with a sign-pretrained or end-to-end-trained visual frontend.

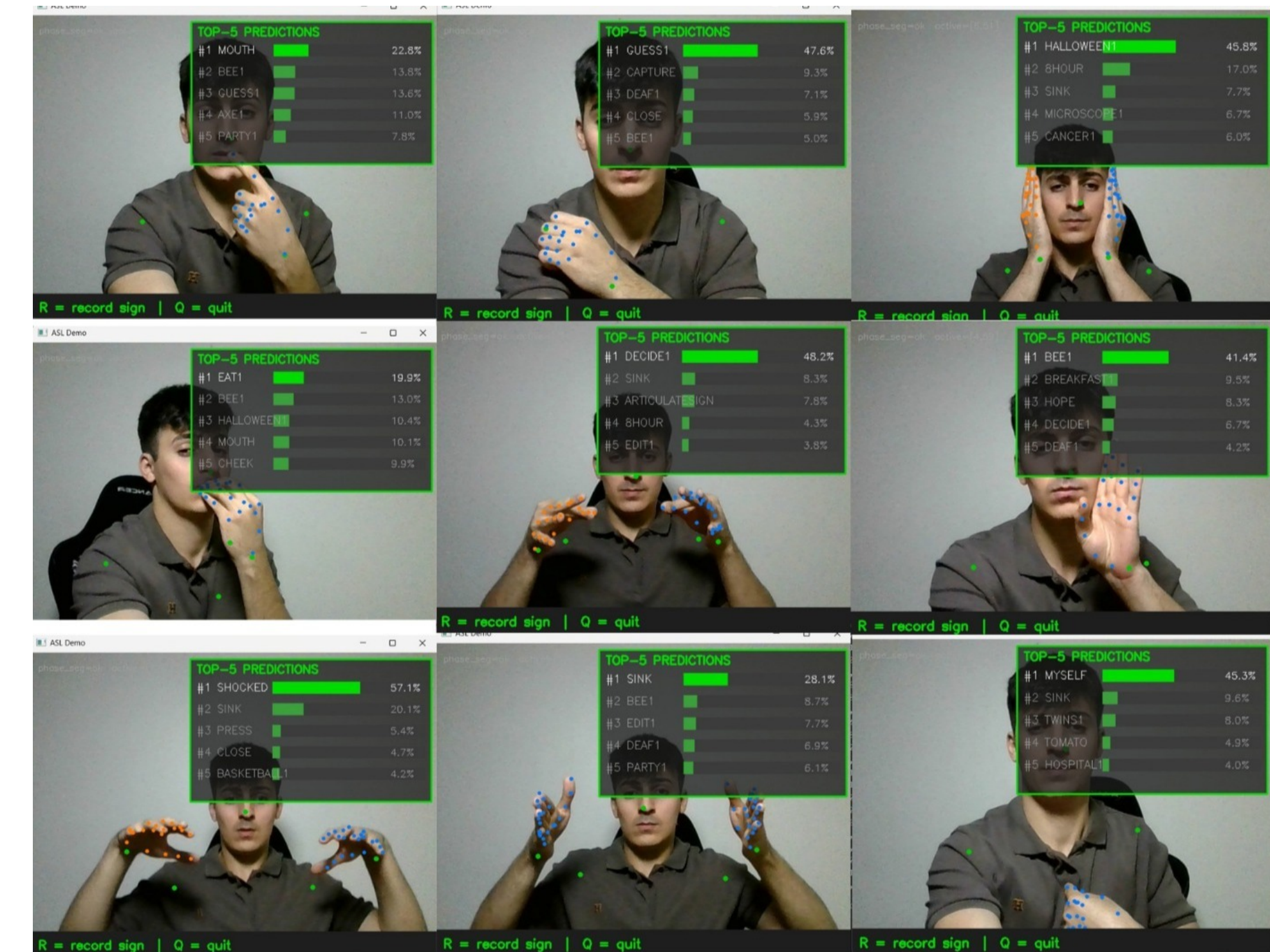


Figure 1:Real-time isolated-sign demonstrator: live top-5 predictions (Part I).